

scan-no-removelast-seqcoll.sh

LVA-054 JDK21 SequencedCollection guard — a \$6.J anti-bluff source scanner.

Script: `scripts/scan-no-removelast-seqcoll.sh`

Purpose

Enforces that no production Kotlin uses the SequencedCollection-shaped List/MutableList methods that crash on Android runtimes below API 35.

The crash class (LVA-053 / LVA-054)

Kotlin MutableList.removeLast() / removeFirst() and List.getFirst() / getLast() compile, under JDK21 + AGP core-library-desugaring, to the JDK21 java.util.SequencedCollection methods (java.util.List.removeLast, etc.). Those methods do NOT exist on the Android platform's java.util.ArrayList below API 35, so a real user on Android 14 / 13 / 12 hits a runtime NoSuchElementException. JVM unit tests pass (desktop ArrayList HAS the methods) — exactly the bluff this gate evicts: the test is green, the feature crashes for the user.

What it does

1. Greps tracked production Kotlin (*.kt, excluding test sources) for the forbidden removeLast() / removeFirst() / getFirst() / getLast() call shapes on List/MutableList receivers.
2. Exits 1 (printing the offending file:line) on any match; exit 0 when clean.

Use .removeAt(lastIndex) / .first() / .last() (the Kotlin-stdlib forms that do NOT desugar to SequencedCollection) instead.

Usage

`bash scripts/scan-no-removelast-seqcoll.sh`

Wiring

Invoked by scripts/check-constitution.sh (pre-push gate);
hermetic test at tests/check-constitution/
test_no_remove_last_seqcoll.sh.

Constitutional bindings

§6.J (tests/gates must guarantee the product works for real users
— this gate catches a runtime crash class invisible to JVM unit
tests), §6.R-adjacent (mechanical source scanner).